SEMESTER EXAMINATION-MAY 2023
CLASS: B.TECH SEMESTER: VI
PAPER CODE: BET-C610
PAPER TITLE: EMBEDDED SYSTEM

Time: 3 hour Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any ***five*** questions in about 150 words each. Each question carries six marks. (5 X 6 = 30 Marks)

Question-1: What is an embedded system? How does it differ from a general-purpose computer system?

Question-2: What is a Real-Time Operating System (RTOS)? How is it different from a general-purpose operating system?

Question-3: Discuss the use of watch dog timer in embedded system design.

Question-4: Explain Embedded Design life cycle with diagram.

Question-5: Differentiate CISC and RISC processor.

Question-6: Define tasks, states and semaphores.

Question-7: Explain TMOD and TCON register of 8051 microcontroller.

Question-8: Write a program in C language to interface 8 LED's with microcontroller 8051 on any port of microcontroller 8051 / 89c51.

Question-9: Draw and compare von-Neumann and Harvard architecture.

Question-10: Explain I2C bus frame format.

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any ***four*** questions in detail. Each question carries 10 marks. (4 X 10 = 40 Marks)

Question-11: Explain the design parameters of embedded system and discuss its significance.

Question-12: What do you mean by hardware and software partitioning and co-design, discuss in detail.

Question-13: Explain in brief about the following. (i) FSMD (ii) FSM (iii) data-path (iv) ASIP

Question-14: Draw and explain the architecture of 80386 processor.

Question-15: (a) Discuss the memory organization of 8051 Microcontroller. (b) Discuss Addressing modes in 8051 microcontroller.

Question-16: A switch is connected to pin P1.5 (8051 microcontroller). Write a program to check the status of the switch and make the following decision.

a) if SW=0, send '1' to P2

b) if SW=1 send '0' to P2

Question-17: Draw the interfacing circuit and write a program to interface LCD (Liquid Crystal Display unit) with microcontroller 8051.

Question-18: Write short notes on following:

(a) I/O sinking and Sourcing

(b) Direct memory access